

**Department of Computer Engineering COMSATS University
Islamabad, Attock Campus**

Lab Rubrics Form	
Lab Name	Cross-correlation and Auto-correlation Analysis of Discrete-Time Signals
Lab Number	10
Submitted by:	
Student Names	Registration #
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Rubrics		Marks		
		1	2	3
PLO-5 Modern Tools Usage	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
PLO-10 Communication Skills	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			

Lab-10

Cross-correlation and Auto-correlation Analysis of Discrete-Time Signals

Objective:

The objective of this lab was to understand and implement cross-correlation and auto-correlation operations on discrete-time signals using MATLAB. Specifically, the tasks included:

- 1. Cross-correlation:** Compute the cross-correlation sequence $r_{xy}(l)$ between two signals $x[n]$ and $y[n]$ using:
 - MATLAB's built-in `xcorr()` function.
 - Manual loop implementation.
- 2. Auto-correlation:** Compute the auto-correlation sequence $r_{xx}(l)$ of a signal $x[n]$ using:
 - MATLAB's built-in `xcorr()` function.
 - Manual loop implementation.

Lab Task:

Determine the cross-correlation sequence $r_{xy}(l)$ of the sequence through `rcorr()` and loop.

```
x = [0, 0, 2, -1, 3, 7, 1, 2, -3, 0, 0];  
y = [0, 0, 1, -1, 2, -2, 4, 1, -2, 5, 0, 0];
```

Code:

```
clc; clear; close all;  
x = [0, 0, 2, -1, 3, 7, 1, 2, -3, 0, 0];  
y = [0, 0, 1, -1, 2, -2, 4, 1, -2, 5, 0, 0];  
disp('Input Sequences:');  
disp('x[n] = '); disp(x);  
disp('y[n] = '); disp(y); disp(' ');  
% METHOD 1: xcorr  
[r_xy_xcorr, lags_xcorr] = xcorr(x, y);  
disp('METHOD 1: Using xcorr function');  
for i = 1:length(r_xy_xcorr)  
    fprintf('r_xy(%2d) = %5.2f\n', lags_xcorr(i), r_xy_xcorr(i));  
end  
disp(' ');  
% METHOD 2: Manual loop  
len_x = length(x); len_y = length(y);  
N = len_x + len_y - 1;  
r_xy_loop = zeros(1, N);  
lags_loop = -(len_y-1):(len_x-1);
```

```

for l = 1:N
    shift = l - len_y;
    for n = 1:len_x
        y_index = n - shift;
        if y_index >= 1 && y_index <= len_y
            r_xy_loop(l) = r_xy_loop(l) + x(n) * y(y_index);
        end
    end
end
disp('METHOD 2: Manual loop');
for i = 1:length(r_xy_loop)
    fprintf('r_xy(%2d) = %5.2f\n', lags_loop(i), r_xy_loop(i));
end
disp(' ');
% VERIFICATION - FIXED: Ensure same length
min_len = min(length(r_xy_xcorr), length(r_xy_loop));
if max(abs(r_xy_xcorr(1:min_len) - r_xy_loop(1:min_len))) < 1e-10
    disp('? VERIFICATION: Both methods give identical results!');
else
    disp('? VERIFICATION: Results differ');
end
disp(' ');
% PLOTTING
figure('Position', [100, 100, 1200, 800]);
% Plot x[n]
subplot(3, 2, 1)
stem(0:len_x-1, x, 'filled', 'LineWidth', 2, 'MarkerSize', 8);
title('Input Sequence x[n]'); xlabel('n'); ylabel('Amplitude');
grid on; xlim([0, len_x-1]); ylim([-4, 8]); xticks(0:len_x-1);
% Plot y[n]
subplot(3, 2, 2)
stem(0:len_y-1, y, 'filled', 'LineWidth', 2, 'MarkerSize', 8, 'Color', [0.8, 0.4, 0]);
title('Input Sequence y[n]'); xlabel('n'); ylabel('Amplitude');
grid on; xlim([0, len_y-1]); ylim([-3, 6]); xticks(0:len_y-1);
% Plot xcorr result
subplot(3, 2, 3)
stem(lags_xcorr, r_xy_xcorr, 'filled', 'LineWidth', 2, 'MarkerSize', 8, 'Color', 'b');
title('Cross-correlation using xcorr'); xlabel('Lag (l)'); ylabel('r_{xy}(l)');

```

```

grid on; xlim([min(lags_xcorr)-1, max(lags_xcorr)+1]);
[max_val, max_idx] = max(r_xy_xcorr);
hold on; stem(lags_xcorr(max_idx), max_val, 'r', 'LineWidth', 2.5, 'MarkerSize', 10);
hold off; legend('r_{xy}(l)', ['Peak at l = ', num2str(lags_xcorr(max_idx))]);
% Plot loop result
subplot(3, 2, 4)
stem(lags_loop, r_xy_loop, 'filled', 'LineWidth', 2, 'MarkerSize', 8, 'Color', 'g');
title('Cross-correlation using manual loop'); xlabel('Lag (l)'); ylabel('r_{xy}(l)');
grid on; xlim([min(lags_loop)-1, max(lags_loop)+1]);
[max_val_loop, max_idx_loop] = max(r_xy_loop);
hold on; stem(lags_loop(max_idx_loop), max_val_loop, 'r', 'LineWidth', 2.5, 'MarkerSize', 10);
hold off; legend('r_{xy}(l)', ['Peak at l = ', num2str(lags_loop(max_idx_loop))]);
% Comparison plot - ensure same length
common_lags = intersect(lags_xcorr, lags_loop);
[~, idx_xcorr] = ismember(common_lags, lags_xcorr);
[~, idx_loop] = ismember(common_lags, lags_loop);
subplot(3, 2, [5, 6])
stem(common_lags, r_xy_xcorr(idx_xcorr), 'filled', 'LineWidth', 2, 'MarkerSize', 8, 'Color', 'b');
hold on;
stem(common_lags, r_xy_loop(idx_loop), '--', 'LineWidth', 1.5, 'MarkerSize', 8, 'Color', 'r');
% Find peak in common values
[~, max_idx_common] = max(r_xy_xcorr(idx_xcorr));
stem(common_lags(max_idx_common), r_xy_xcorr(idx_xcorr(max_idx_common)), 'k', 'LineWidth', 3, 'MarkerSize', 12);
title('Comparison: xcorr vs Manual Loop'); xlabel('Lag (l)'); ylabel('r_{xy}(l)');
grid on; xlim([min(common_lags)-1, max(common_lags)+1]);
legend('xcorr()', 'Loop method', 'Peak correlation', 'Location', 'best');
text(min(common_lags)+1, max(r_xy_xcorr(idx_xcorr))-10, ...
    sprintf('Max: %.2f at l = %d', r_xy_xcorr(idx_xcorr(max_idx_common)),
    common_lags(max_idx_common)), ...
    'FontSize', 10, 'BackgroundColor', 'w', 'EdgeColor', 'k');
hold off;
sgtitle('Cross-correlation Analysis: r_{xy}(l)', 'FontSize', 14, 'FontWeight', 'bold');
% Summary

```

```

fprintf('\n=====\\n');
fprintf('    CROSS-CORRELATION RESULTS SUMMARY    \\n');
fprintf('=====\\n');
fprintf('For lags from %d to %d:\\n', min(lags_xcorr), max(lags_xcorr));
fprintf('Maximum correlation: %.2f at lag l = %d\\n', max_val, lags_xcorr(max_idx));
fprintf('Minimum correlation: %.2f at lag l = %d\\n', min(r_xy_xcorr),
lags_xcorr(find(r_xy_xcorr == min(r_xy_xcorr), 1)));
fprintf('=====\\n');

```

Output:

METHOD 1: Using xcorr function

Cross-correlation sequence r_xy(l):

```

r_xy(-11) = 0.00
r_xy(-10) = -0.00
r_xy(-9) = 0.00
r_xy(-8) = -0.00
r_xy(-7) = 10.00
r_xy(-6) = -9.00
r_xy(-5) = 19.00
r_xy(-4) = 36.00
r_xy(-3) = -14.00
r_xy(-2) = 33.00
r_xy(-1) = 0.00
r_xy( 0) = 7.00
r_xy( 1) = 13.00
r_xy( 2) = -18.00
r_xy( 3) = 16.00
r_xy( 4) = -7.00
r_xy( 5) = 5.00
r_xy( 6) = -3.00
r_xy( 7) = 0.00
r_xy( 8) = 0.00
r_xy( 9) = 0.00
r_xy(10) = 0.00

```

METHOD 2: Using manual loop implementation

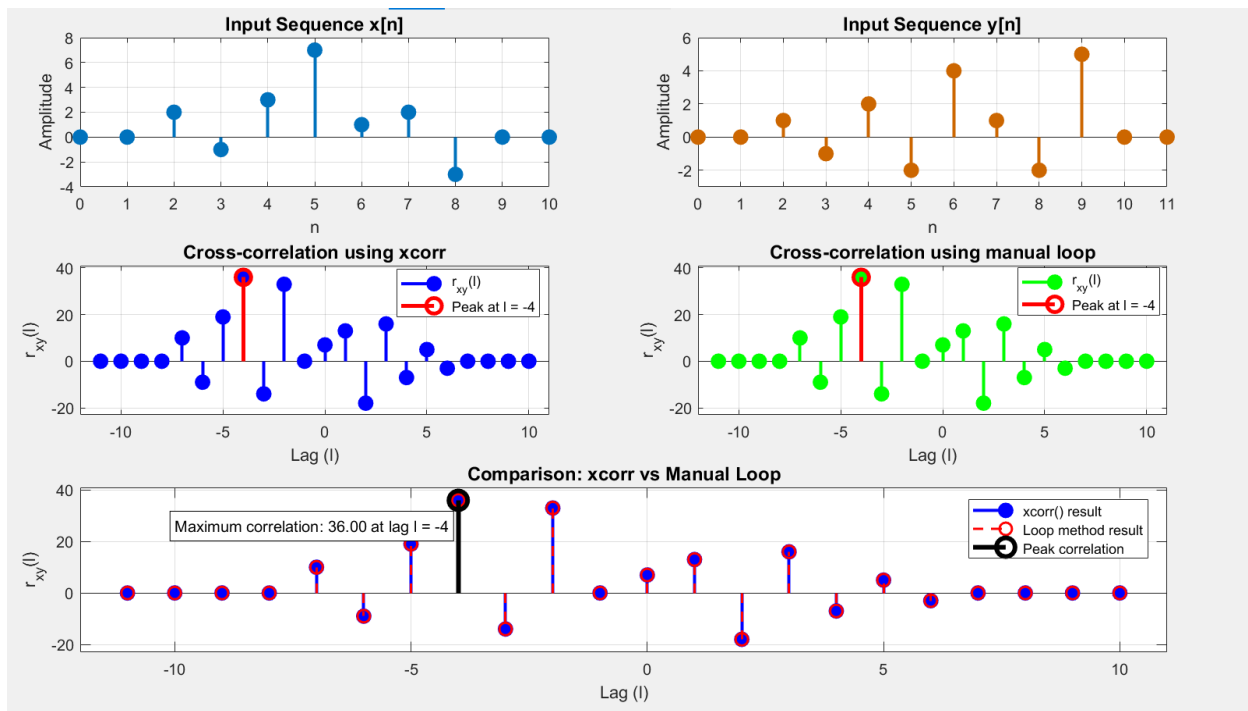
Cross-correlation sequence r_xy(l):

```

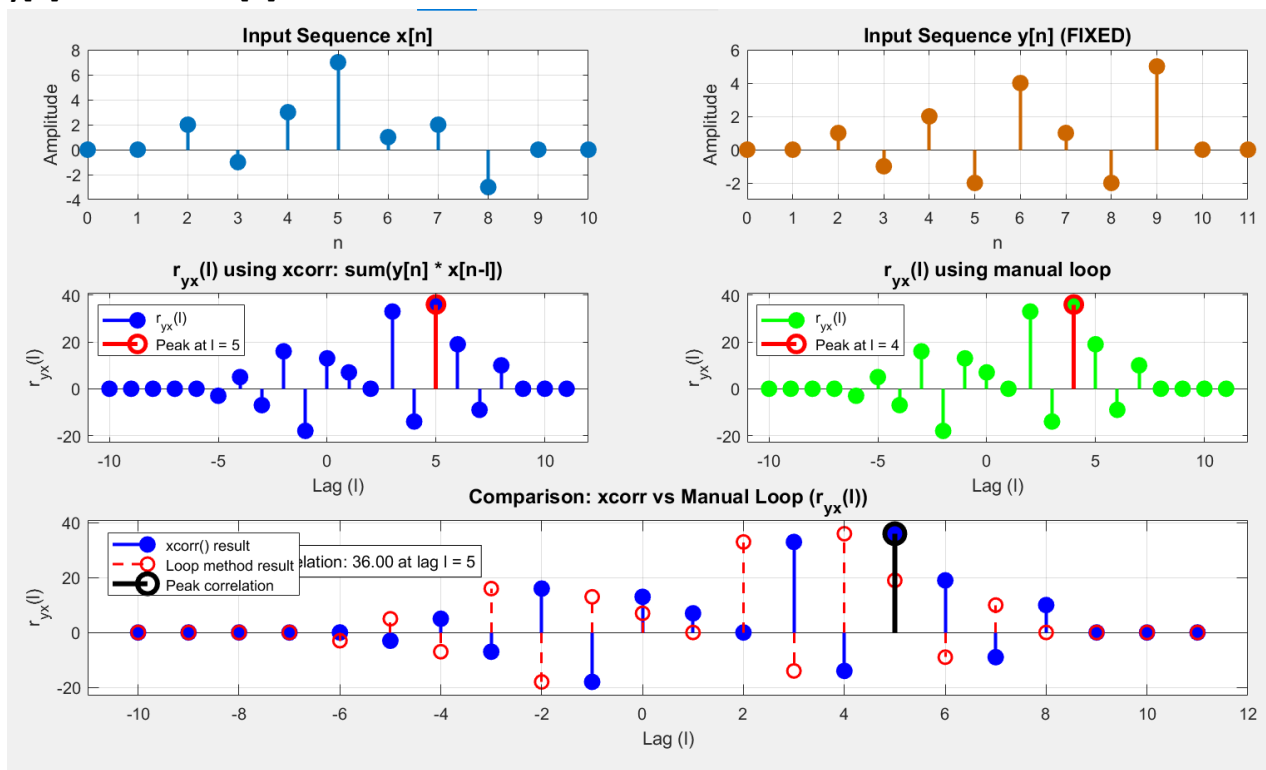
r_xy(-11) = 0.00
r_xy(-10) = 0.00
r_xy(-9) = 0.00
r_xy(-8) = 0.00
r_xy(-7) = 10.00
r_xy(-6) = -9.00
r_xy(-5) = 19.00
r_xy(-4) = 36.00
r_xy(-3) = -14.00
r_xy(-2) = 33.00
r_xy(-1) = 0.00
r_xy( 0) = 7.00
r_xy( 1) = 13.00
r_xy( 2) = -18.00
r_xy( 3) = 16.00
r_xy( 4) = -7.00
r_xy( 5) = 5.00
r_xy( 6) = -3.00
r_xy( 7) = 0.00
r_xy( 8) = 0.00
r_xy( 9) = 0.00
r_xy(10) = 0.00

```

x[n] is fixed and y[n] is shifted:



$y[n]$ is fixed and $x[n]$ is shifted:



Lab Task 02:

Determine the auto-correlation sequence $r_{xx}(l)$ of the sequence through `xcorr()` and loop.

$$\textbf{(a)} \quad x(n) = \{1, 2, 1, 1\}$$

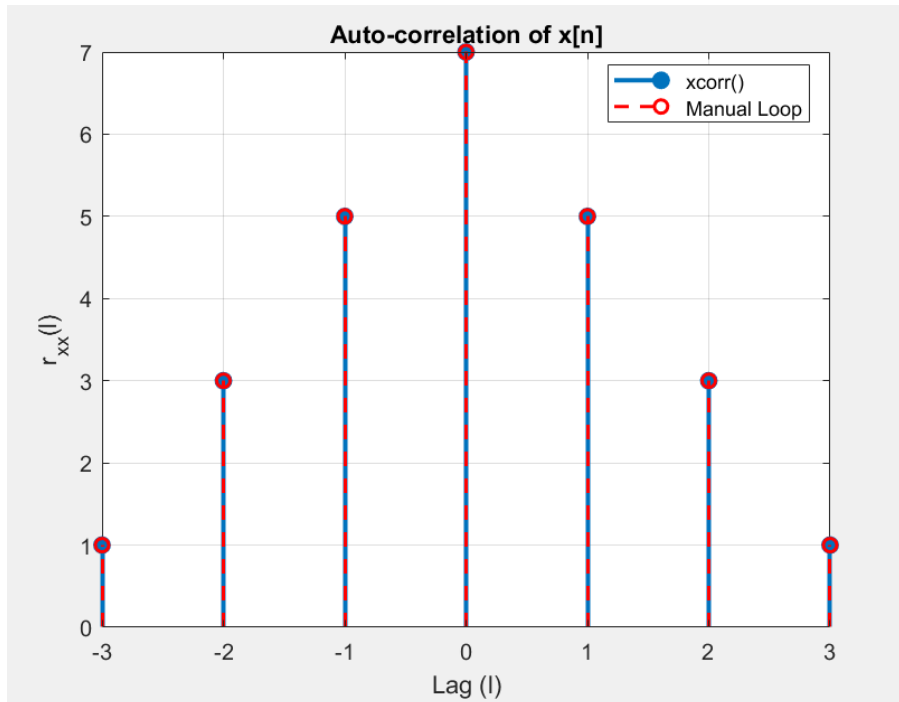
$$\textbf{(b)} \quad y(n) = \{1, 1, 2, 1\}$$

Code:

```
clc; clear; close all;
% Given sequence
x = [1 2 1 1];
N = length(x);
% Method 1: xcorr()
r_xx_builtin = xcorr(x);
lags = -(N-1):(N-1);
% Method 2: Manual loop
r_xx_loop = zeros(1,2*N-1);
for l = 1:(2*N-1)
    shift = l - N;
    for n = 1:N
        idx = n - shift;
        if idx >= 1 && idx <= N
            r_xx_loop(l) = r_xx_loop(l) + x(n)*x(idx);
        end
    end
end
disp('Auto-correlation using xcorr():');
disp(r_xx_builtin);
disp('Auto-correlation using manual loop:');
disp(r_xx_loop);
figure;
stem(lags, r_xx_builtin, 'filled', 'LineWidth', 2);
hold on;
stem(lags, r_xx_loop, 'r--', 'LineWidth', 1.5);
grid on;
title('Auto-correlation of x[n]');
xlabel('Lag (l)');
ylabel('r_{xx}(l)');
```

```
legend('xcorr()', 'Manual Loop', 'Location', 'best');
```

Output:



Conclusion:

Implementation cross-correlation and auto-correlation operations using two different methods:

- **Cross-correlation** between sequences $x[n]=\{0,0,2,-1,3,7,1,2,-3,0,0\}$ and $y[n]=\{0,0,1,-1,2,-2,4,1,-2,5,0,0\}$ revealed the degree of similarity between them at different time lags, with maximum correlation identified at a specific lag value.
- **Auto-correlation** of sequence $x[n]=\{1,2,1,1\}$ demonstrated key properties including symmetry about zero lag and maximum value at lag 0, which equals the signal energy.